

Type: DSC1-5 (Major) B.Sc(ECS)-II (Semester IV) Course Title: Core Java (Paper Code:)		
Credits: Theory – (2)		Practical's – (1)
Total Lectures: 30 Hrs.		Contact Hrs. (L) : 2
University Evaluation: 30 Marks		Internal Evaluation: 20 Marks
Course Outcomes:		
1.	Knowledge of the structure and model of the Java programming language.	
2.	Use the Java programming language for various programming technologies.	
3.	Develop software in the Java programming language.	
4.	Evaluate user requirements for software functionality required to decide whether the Java programming language can meet user requirements.	
Unit	Content	Lect.
I	Introduction to Java : Introduction- History, Features of Java, Installation and configure JDK. JVM architecture, JRE, Structure of java program, Datatypes, variables, identifiers, operators, keywords, java arrays, Control Statement- if, if-else, switch, for loop, while loop, do while loop, nested loop, break, continue, comments, java functions. Object oriented Concepts- class, objects, methods, constructor, naming conventions, static keyword, use of this keyword. Inheritance: Types of inheritance, use of super keyword, Access/Scope specifiers, Method overriding, Method overloading, Abstract classes, Final keyword, Packages and Interfaces: create and import package, Difference between classes and interfaces, runtime polymorphism, Wrapper classes.	15
II	Exception handling: Use of try, catch, finally, throw and throws keywords, Built-in exception classes, creating User defined exceptions, Multithreading: Concept, Life cycle of a thread, Thread class, Runnable interface, Methods in Thread class, sleep, interrupt, join, priority etc., Thread Synchronization, Usage of wait and notify() methods. File IO programming: Concept, Types of file, I/O stream classes, Readers and Writers classes, Console I/O, Object Stream and Serializable interface.	15
Reference Books:		
1.	Core Java by Dr. R. Nageshwar Rao.	
2.	James Gosling, Bill Joy, Guy L Steele Jr, Gilad Bracha, Alex Buckley The Java Language	

	Specification, Java SE 8 Edition (Java Series), Published by Addison Wesley, 2014.
3.	Cay S. Horstmann, Gary Cornell, Core Java 2 Volume 2 - Advanced Features, 9th Edition, Printice Hall. 2013.
4.	Bruce Eckel, Thinking in Java, 3rd Edition, PHI, 2002.
5.	E. Balaguruswamy, Programming with Java, 4th Edition, McGraw Hill. 2009.